

Instrumentation Lecture 02

Instrumentation and control.

Classifications of Instrumentation

- Mechanical instruments
- Electrical instruments
- Electronic instruments

Classification in based on the various effects of Electricity.

- Magnetic effect
- Heating effect
- Chemical effect
- Electrostatic effect
- Electromagnetic induction effect.

Classification based on Nature of their Operation

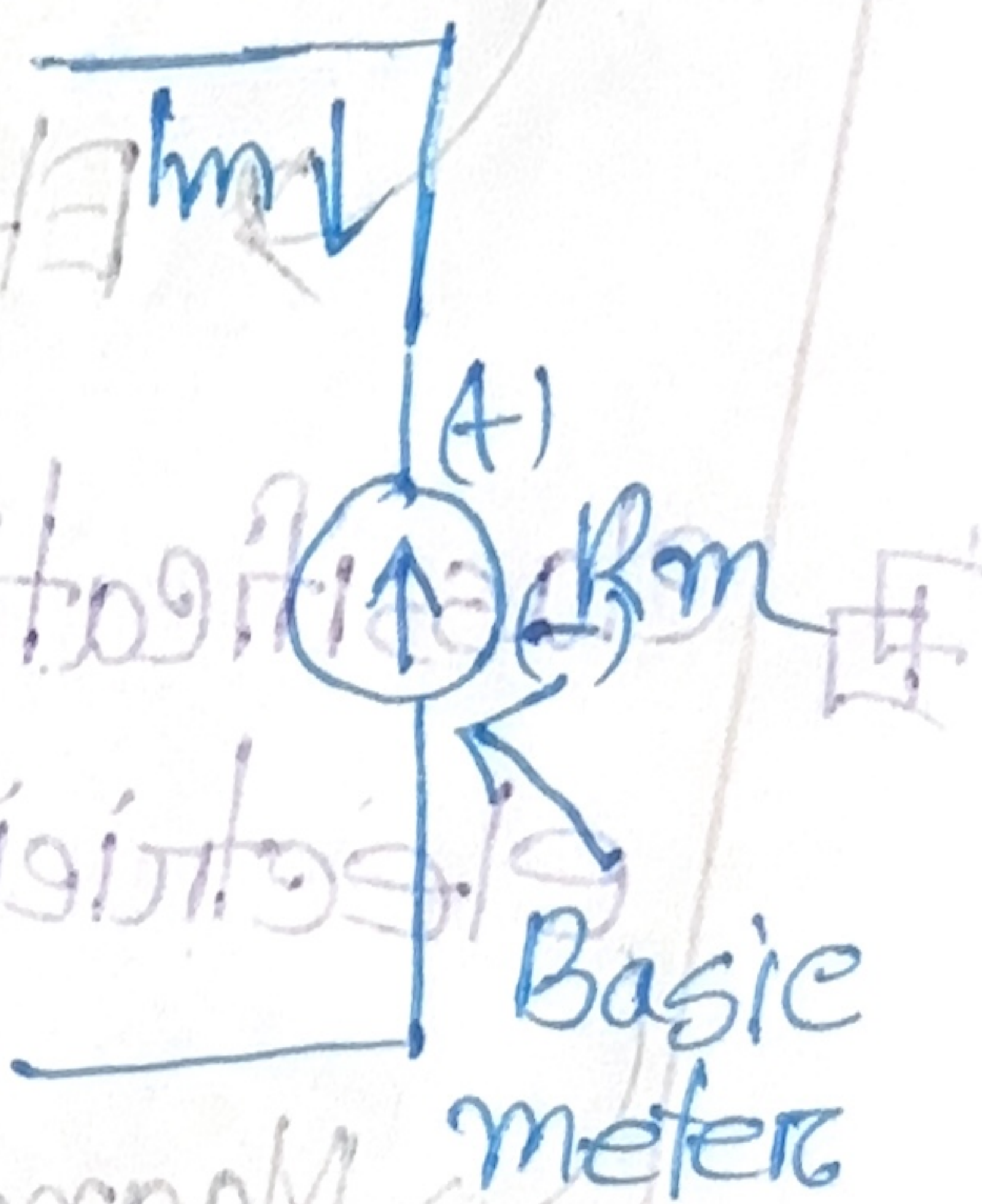
- Indicating instruments
- Recording instruments
- Integrating instruments

Circuit Symbol of basic PMMC Meter.

Here

I_m = full-scale deflection current through the basic meter

R_m = resistance of the basic meter.



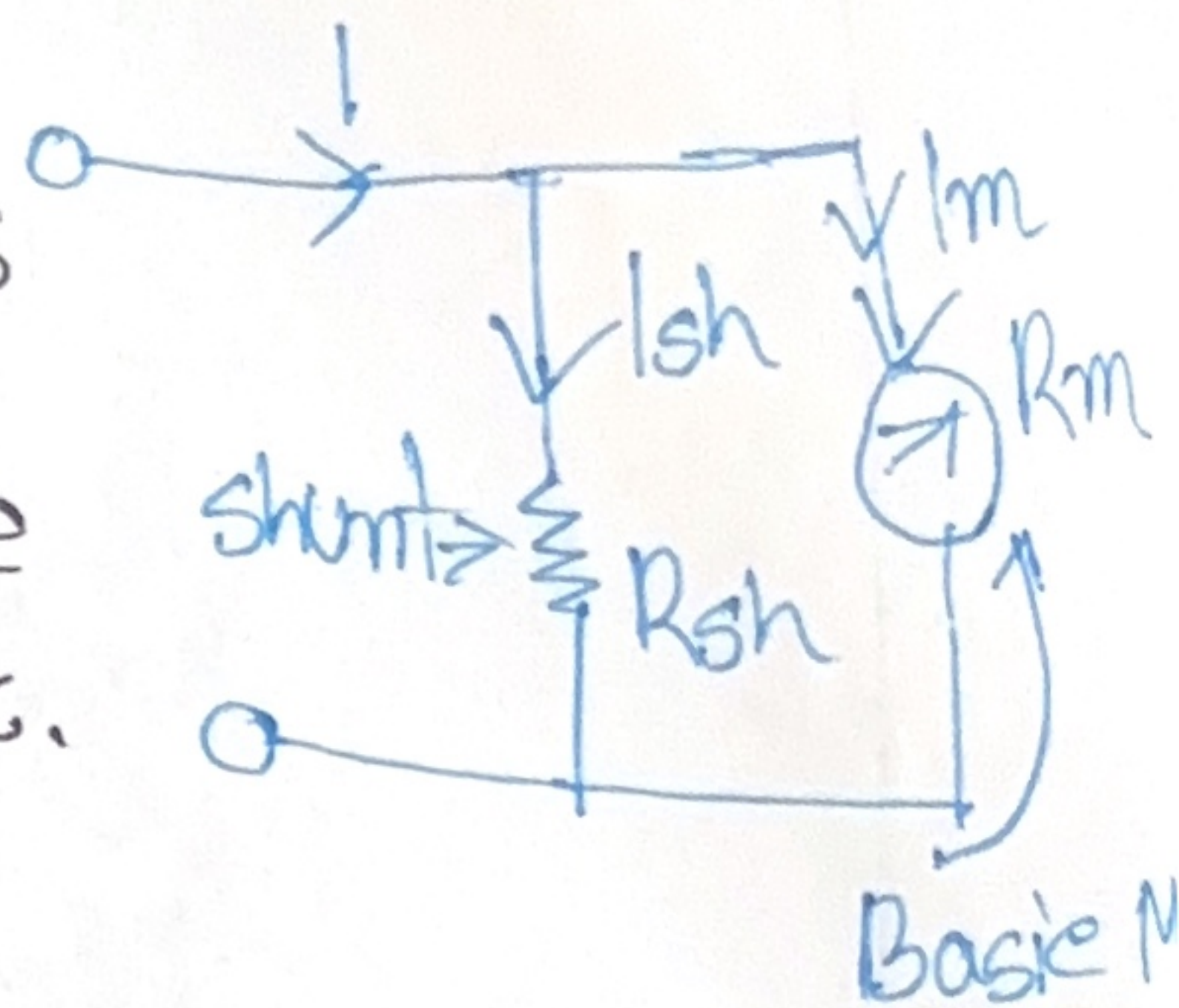
- Electromagnetic induction effect
- Electrostatic effect
- Chemical effect
- Heating effect

R_{sh}

Multirange

Various
can be
suitable
to increase
current range

De PMMC Ammeters.
For large current measurements a shunt resistance R_{sh} must be connected to the basic meter.



$$I_{sh} R_{sh} = I_m R_m$$

$$R_{sh} = \frac{I_m R_m}{I_{sh}}$$

but $I_{sh} = I - I_m$

$$R_{sh} = \frac{I_m R_m}{(I - I_m)}$$

$$R_{sh} = \frac{R_m}{m-1} \text{ where } m = \frac{I}{I_m}$$

Multirange Ammeters.

Various values of resistances can be used by means of suitable switching arrangement to increase the number of current range.

